FBISE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: 2SA-I 2207 — D(6073)

Retyped faithfully from the original board paper — same questions, no changes.

Note: PHYSICS SSC-II. SECTION – A (Marks 12). Time allowed: 15 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 6073. Fill the relevant bubble against each question:

Q1. The most suitable means of reliable continuous communication through air by using a cell phone is:

(A) Infrared waves
(B) Ultraviolet waves
(C) Microwaves
(D) Radio waves

سیل فون کے استعمال میں ہوا کے ذریعے قابل بھروسہ کمیونیکیشن کا مناسب ترین ذریعہ ____ ہے۔

(A) انفراریڈ ویوز
(B) الٹراوائلٹ ویوز
(C) مائیکرو ویوز
(D) ریڈیو ویوز

Q2. Complete the following nuclear reaction. ${}^{a}ZX \rightarrow$ ____ + β^{-} + Energy

(A) ${}^{a-2}Z_{+1}Y$
(B) ${}^{a}YZ_{+1}$
(C) ${}^{a-4}Z_{+1}Y$
(D) ${}^{a-4}Z_{-2}Y$

درج ذیل نیوکلیائی تعامل کو مکمل کیجیے۔ ${}^{-a}ZX \rightarrow$ ____ + β + انرجی

Q3. During SHM an object has ____ at mean position.

(A) P.E = 0, K.E = 0
(B) P.E = K.E
(C) P.E = max, K.E = 0
(D) P.E = 0, K.E = max

سمپل ہارمونک موشن کے دوران ایک جسم مڈل پوزیشن پر ____ کا حامل ہوتا ہے۔

Q4. The radio waves of wavelength 3.33m are transmitted by an FM station a frequency of: whereas V=speed of light

(A) 0.01×10^{-6} Hz
(B) 0.9 Hz
(C) 9.99×10^{8} Hz
(D) 90×10^{6} Hz

ایک FM اسٹیشن 3.33m ویو لینتھ کی ریڈیو ویوز ____ کی فریکوئنسی پر چل رہا ہے جبکہ V=روشنی کی رفتار

Q5. The intensity level of 80 db corresponds to sound intensity of ____ where Intensity level = $10\log(I/I_0)$

(A) 10^{-20} Wm^{-2}
(B) 10^{-4} Wm^{-2}
(C) 10^{68} Wm^{-2}
(D) 10^{-92} Wm^{-2}

80 db کی انٹینسٹی لیول ____ ساؤنڈ انٹینسٹی کے مساوی ہے۔ جبکہ انٹینسٹی لیول $=10\log(I/I_0)$

Q6. Total internal reflection occurs when.

(A) $\theta_i > \theta c$
(B) $\theta_i \leq \theta c$
(C) $\theta_i = \theta c$
(D) $\theta_i < \theta c$

ٹوٹل انٹرنل ریفلیکشن کب واقع ہوتا ہے؟

Q7. A convex lens of focal length 6.00cm has the power:

(A) 0.167 D
(B) 0.016 D
(C) 16.70 D
(D) 1.67 D

ایک 6.00cm فوکل لینتھ کے کنویکس لینز کا پاور ____ ہے:

Q8. The SI unit of capacitance of an electrolytic capacitor is:

(A) Volt
(B) Ampere
(C) Farad
(D) Coulomb

ایک الیکٹرولائیٹک کیپیسٹر کی کیپیسٹنس کا SI یونٹ ____ ہے۔

(A) وولٹ
(B) ایمپیئر
(C) فیراڈ
(D) کولمب

Q9. Two small charged spheres are separated by 2mm. Which of the following would produce the greatest attractive force?

(A) +3q and −1q
(B) +2q and −2q
(C) +1q and +4q
(D) −1q and −4q

دو چھوٹے چارجڈ سفیرز کو 2mm کے فاصلے پر رکھا گیا ہے۔ درج شدہ میں سے کس انتخاب کے لیے سب سے زیادہ کشش کی قوت ہوگی؟

Q10. If two resistors $R_1 = 2\Omega$ and $R_2 = 2\Omega$ are connected in parallel across a voltage source of 6V then equivalent resistance (Re) of the circuit is:

(A) 1Ω (B) 2Ω

(C) ½Ω (D) 4Ω

اگر دو رزسٹرز $R_1 = 2\Omega$ اور $R_2 = 2\Omega$ کو 6V کے وولٹیج سورس کے ساتھ پیرالل میں جوڑے جائیں تو سرکٹ کی مساوی رزسٹنس (Re) ____ ہے:

Q11. Magnetic force on a current carry in conductor is increased if:

(A) Length of wire is decreased (B) Strength of magnetic field is decreased

(C) The current is decreased (D) The current is increased

ایک کرنٹ دارہ راہ پر میگنیٹک فورس بڑھنے کی وجہ:

(A) تار کی لمبائی کو کم کیا جائے **(B)** میگنیٹک فیلڈ کی شدت کو کم کر دیا جائے

(C) کرنٹ کو کم کیا جائے **(D)** کرنٹ کو بڑھایا جائے

Q12. For AND gate output X = 1 when inputs A and B are. (Note: an AND gate diagram with inputs A, B and output X is given.)

(A) 1,1 (B) 0,0

(C) 1,0 (D) 0,1

انپٹ (AND) گیٹ کے لیے آؤٹ پٹ $X = 1$ کے برابر ان پٹ A اور B کی کس صورت میں ہوگی؟

> Note: Formulas given on Section-A: $v = f\lambda$; Intensity level = $10\log(I/I_0)$; $P = 1/f$; $F = 1/(4\pi\varepsilon_0) \times q_1q_2/r^2$ or $F = K\, q_1q_2/r^2$; $1/Req = 1/R_1 + 1/R_2$; A.B = X; Speed of light = 3×10^8 ms^{-1}. —2SA-I 2207(HA)-6073 —. Page 1 of 2. Page 2 of 2.

> Note: PHYSICS SSC–II. Time allowed: 2:45 Hours. Total Marks Sections B and C: 53. NOTE: Answer any eleven parts from Section 'B' and attempt any two questions from Section 'C' on the separately provided answer book. Write your answers neatly and legibly. (The source also carries the full Urdu translation of this paper.)

Q. SECTION – B (Marks 33)

Q. 2 Answer any ELEVEN parts from the following. All parts carry equal marks. (11 x 3 = 33)

(i) Discuss and show that damping progressively reduces the amplitude of oscilation.

(ii) Find the time period of a simple pendulum of 1m length, placed on moon. The value of 'g' on the surface of moon is 1/6 of its value on earth i.e. $g_m = g_e/6$

(iii) Discuss the importance of acoustics in architectural design of lecture halls and theatre halls.

(iv) What is Cathode Ray Oscilloscope CRO? Make a list of its use.

(v) A ship is anchored where the depth of water is 120m. An ultrasonic signal sent to the bottom of the lake returns in 0.16s. What is the speed of sound in water?

(vi) Describe the use of logic gates in house safety alarm.

(vii) Draw neat V-I graphs to show the voltage-current relationship in metals, filament lamp and thermistor.

(viii) State Coulomb's law and give its mathematical relation. Write the value and SI units of constant of proportionality 'K' in air.

(ix) Find the value of Equivalent resistance Re in the parallel combination of three resistors $R_1 = 18\Omega$, $R_2 = 9\Omega$ and $R_3 = 6\Omega$.

(x) How can the direction of a magnetic field formed around a current carrying conductor be determined?

(xi) A step-down transformer has a turn ration of 100:1. An AC voltage of amplitude 170V is applied to the primary. What is the voltage in the secondary?

(xii) Show that the circuit given below in figure acts as OR gate. Write down the related Boolean equations also. (Note: A and B feed a NAND gate followed by a NOT gate, output Y.)

(xiii) What are cathode rays? Explain the deflection of cathode rays by a magnetic field.

(xiv) State any three (03) risks of ICT to society.

(xv) Ashes from a camp fire deep in a cave show carbon-14 activity of only one-eighth the activity of fresh wood. How long ago was that comp fire made? The half-life of Carbon-14 is 5730 years.

Q. SECTION – C (Marks 20)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 10 = 20)

Q. 3 a. Explain in detail the series combination of capacitors. (6) b. What is electromagnetic induction? List any two factors affecting the magnitude of induced emf. Also state Lenz's law and its significance (4)

Q. 4 a. Explain the construction and working of a compound microscope by the help of a ray diagram. Write down the formula to determine the magnification of microscope. (6) b. A wooden bar vibrating on the water surface in a ripple tank has a frequency of 12Hz. The resulting wave has a wavelength of 3cm. What is the speed of the wave. (4)

Q. 5 a. What is radio activity. What is meant by half-life of a radioactive element? Explain how can the half-life of an element be measured. Draw neat graph to illustrate the answer. (6) b. By applying a potential difference of 10V across a conductor a current of 1.5A passes

through it. How much energy would be obtained from the current in 2 minutes according to Joule's law? (4)

> Note: Formulas given on subjective paper: $T = 1/(2\pi)\sqrt{(l/g)}$; $S = vt$; $1/Req = 1/R_1 + 1/R_2 + 1/R_3$; $Ns/Np = Vs/Vp$; $A.B = X$; $A = \bar{A}$; $X = A+B$; $E = VIt$; $v = f\lambda$; $N = N_0/8$; $N = N_0 \times 1/2^n$; Total Time = n × Half lives. —2SA-I 2207 (HA)—